

# Project Documentation: Position Salary Prediction using Simple Linear Regression

## 1. Executive Summary

This project demonstrates the implementation of a **Simple Linear Regression** model using Python, `pandas`, `scikit-learn`, and `matplotlib`. The primary goal is to model and predict employee salaries based on their job level/position within an organization.

## 2. Dataset Architecture

### Dataset Overview

The dataset contains corporate job positions, corresponding hierarchy levels, and annual compensation.

| Column Name | Data Type       | Role            | Description                          |
|-------------|-----------------|-----------------|--------------------------------------|
| Position    | String / Object | Categorical     | Job title/role                       |
| Level       | Integer         | Feature ( $X$ ) | Numerical hierarchy level of the job |
| Salary      | Integer / Float | Target ( $y$ )  | Annual compensation in USD (\$)      |

### Data Preview

| Index | Position         | Level ( $X$ ) | Salary ( $y$ ) |
|-------|------------------|---------------|----------------|
| 0     | Business Analyst | 1             | \$45,000       |

|   |                |   |          |
|---|----------------|---|----------|
| 1 | Junior C       | 2 | \$50,000 |
| 2 | Senior C       | 3 | \$55,000 |
| 3 | Manager        | 4 | \$65,000 |
| 4 | Contry Manager | 5 | \$75,000 |

### 3. Mathematical Model & Methodology

Simple Linear Regression establishes a linear relationship between an independent variable  $X$  (**Level**) and a dependent target variable  $y$  (**Salary**):

$$y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- $y$ : Predicted Salary
- $X$ : Position Level
- $\beta_1$ : Slope (Coefficient)
- $\beta_0$ :  $y$ -intercept
- $\epsilon$ : Residual Error term

### 4. Source Code Execution & Pipeline

Python

# Step 1: Import core dependencies

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
import pandas as pd
```

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

# Step 2: Load the dataset

```
df = pd.read_csv(
```

```
    "/kaggle/input/datasets/farhankarim11/data-set/Position_Salary_Data.csv"
```

```
)
```

# Step 3: Feature & Target Isolation

```
X = df[["Level"]].values
```

```
y = df["Salary"].values
```

```

# Step 4: Model Training
model = LinearRegression()
model.fit(X, y)

# Step 5: Prediction
y_pred = model.predict(X)

# Step 6: Model Evaluation
r2 = r2_score(y, y_pred)
rmse = np.sqrt(mean_squared_error(y, y_pred))

# Model Parameters
print(f"Slope (Coefficient): {model.coef_[0]:.2f}")
print(f"Intercept: {model.intercept_:.2f}")
print(f"R2 Score: {r2:.4f}")
print(f"RMSE: {rmse:.2f}")

```

## 5. Model Evaluation & Performance Analysis

### Executed Output Parameters

- **Slope ( $\beta_1$ ): 49696.97** (Note: Notebook log reflects dataset output with 10 data points)
- **Intercept ( $\beta_0$ ): -94333.33**
- **$R^2$  Score: 0.8250**
- **Root Mean Squared Error (RMSE): \$65,751.37**

### Evaluation Metrics Breakdown

1. **Coefficient (49,696.97):** On average, each single step increase in position level results in an approximate salary increase of **\$49,696.97**.
2.  **$R^2$  Score (0.8250):** Approximately **82.5%** of the variance in salary can be explained linearly by the position level.
3. **RMSE (65,751.37):** Indicates the average magnitude of prediction error across all levels.

## 6. Visualization Analysis

```

Python
# Data Visualization
plt.figure(figsize=(8, 5))
plt.scatter(X, y, color="red", label="Actual Data")
plt.plot(
    X, y_pred, color="blue", linewidth=2, label="Linear Regression Line"
)

```

```
plt.title("Position Level vs Salary (Simple Linear Regression)")
plt.xlabel("Position Level")
plt.ylabel("Salary ($)")
plt.legend()
plt.grid(True)
plt.show()
```

- **Actual Data Points (Red):** Display the ground truth salary values corresponding to each job level.
- **Regression Line (Blue):** Represents the best-fit linear prediction path determined by minimizing the sum of squared residuals.

## 7. Key Findings & Recommendations

- **Linearity Assessment:** While Simple Linear Regression provides a strong initial baseline ( $R^2 = 0.825$ ), executive salaries often scale exponentially at higher position levels.
- **Next Steps for Optimization:**
  1. **Polynomial Regression:** Fitting a polynomial curve ( $y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots$ ) can better capture non-linear wage acceleration at executive levels.
  2. **Feature Expansion:** Adding organizational features (e.g., years of experience, department, region) to build a **Multiple Linear Regression** model.